Foundation:

Fantasy Points Proposed Calculation Metrics:

- For each tournament a player participates in, combine tournament and per game metrics to determine impact on overall player fantasy score
- Fantasy scores reset per season (season defined by fide circuit as jan-december)
- Users can compete in season by season or per tournament leagues

Tournament fantasy metrics :

Scalar constant (default # to scale total per game points, changeable by league creator in settings)

- Prestige (defined by prize money) - (total tournament prize fund/10,000)

Per game fantasy metrics (ordered by weight/priority):

Scalar Constant (default # to scale game metrics, changeable by league creator in settings)

- Time control (scale all factors below - much more points for classical vs rapid and blitz - use ratio of time using base - ignore delay and increment for now. # of minutes/5: - ask ella)

Variable Proportions (default #s out of 100, weights changeable by league creator in settings)

1. Rating gain/loss (captures upsets and non-upsets, wins and losses) (assuming a max gain/loss of 20) (60/100) - a gain of 20 points gets 60 a loss of 20 points gets 0
2. Quality/Sharpe Ratio (lol) of the game using centipawn loss/gain of each move as returns and dividing by vol of these returns (30/100) - currently doing analysis to determine what classifies as a good or bad sharpe ratio in this context, for now let's have a sharpe ratio of 1 is gives 30 points and -1 gives 0 points and so on. No need to code the calculation of this - have done it in another program simply add an input for sharpe ratio and translate to points
    a. Benefits: values positional games more than tactical for both losers and winners
    b. Risks: weighs every move the same - solution is either remove or weigh less:
        i. literal "only moves" (moving out of check) (identify through 0.0 centipawn change)
        ii. technical "only moves" (move that doesn't blunder) (also identify through 0.0 centipawn change)
        iii. theory/book moves (create own identification algo - opening moves seen in 10+ previous games in chessbase or for now, simply identify through chess.com)
        iv. moves made with very little time on the clock (weight can go down exponentially along with time decrease - very low time weigh nearly nothing)
3. The black/white factor (6/100) win with black -6/6, win with white 4/6, lose with white 0/6, lose with black 2/6

4. Psychological Factor (result minus previous ~10 game avg result - max would be 1-0 =1 and min is 0-0=0) (4/100)
5. Future idea to explore, not now: Expecting to under or overperform vs that particular player based on playstyle (opening/middlegame/endgame player) (ex. players with different playstyles more likely to have a decisive result so award/take away less points for a win/loss)